DIGITAL EGYPT PIONEERS INITIATIVE (DEPI) — ROUND 5

SentiLingo

Multilingual Customer Sentiment Analysis Platform



Project Documentation

Part 1 Project Planning & Management

Part 2 Literature Review

Part 3 Requirements Gathering

Track	Microsoft Machine Learning
Group	To be added
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GitHub repository	https://github.com/AhmedMoenes/SentiLingo
Version	1.0 — 17 October 2026

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Document control

Document	SentiLingo Project Documentation — Parts 1–3 · version 1.0 · 17 October 2026 (deadline 19 October 2026)
Prepared by	SentiLingo project team
Next revision	Remaining documentation parts, following the DEPI *Project Documentation Guidelines* schedule

PART 1

1. Project Planning & Management

What SentiLingo is, what it will deliver and when, how the team works, which risks matter most and how success is measured.

1.1 Project Overview

Companies receive customer feedback through app and e-commerce reviews, social media, surveys and support tickets. In Egypt this feedback is written in Modern Standard Arabic (MSA), Egyptian Arabic, English and Franco-Arabic (Arabic in Latin letters, e.g. `el delivery kan 3ady`), often mixed in one message. Tagging it by hand is slow and inconsistent, and many ready-made tools are tuned for English or MSA (see Part 2).

> **Problem statement (from the project brief).** A company receives customer feedback in natural-language text and needs to automatically identify whether feedback expresses positive, negative, or other predefined sentiment categories.

SentiLingo labels each piece of feedback **Negative**, **Neutral** or **Positive** with a confidence score. It compares three model families on the same data splits: a classical baseline (TF-IDF with word and character n-grams, Naive Bayes and Logistic Regression), a BiLSTM, and a fine-tuned pretrained Transformer. All experiments are tracked in MLflow. The best model is chosen by macro-F1 on a frozen test set, its errors are analysed, and it can optionally be served from Azure Machine Learning.

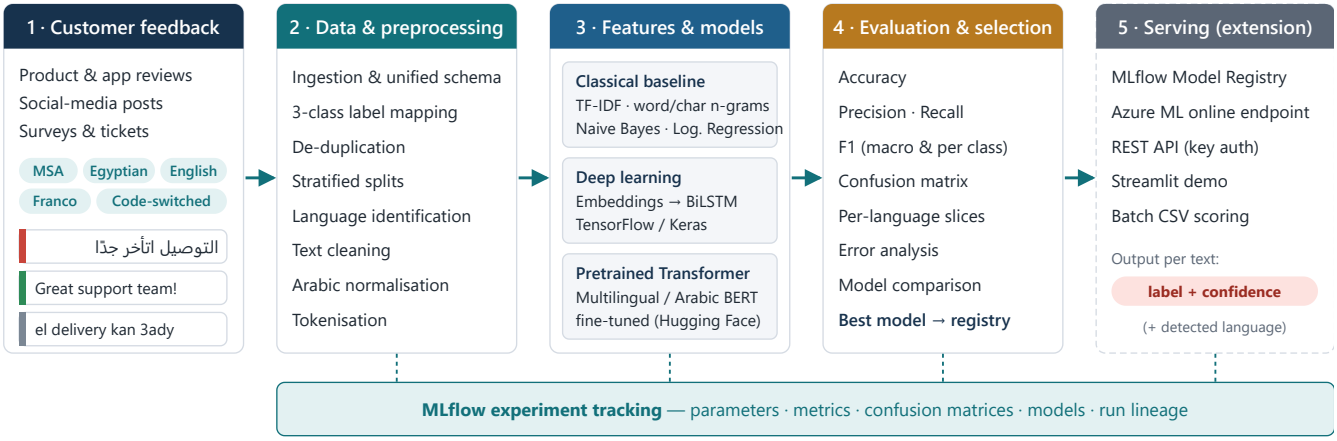


Figure 1 — SentiLingo solution overview.

1.2 Objectives

ID	Objective	Due (2026)
O1	Build a unified Arabic (MSA and Egyptian) and English dataset with three sentiment classes, frozen splits and a data card.	25 Oct
O2	Build a configurable preprocessing pipeline: cleaning, Arabic normalisation, tokenisation.	1 Nov
O3	Train classical baselines: TF-IDF with n-grams, plus Naive Bayes and Logistic Regression.	1 Nov
O4	Train a BiLSTM and fine-tune at least one pretrained Transformer.	15 Nov
O5	Compare all models, analyse errors and register the best model in MLflow.	22 Nov
O6	*(Optional)* Deploy the best model to an Azure ML endpoint with a small demo app.	29 Nov
O7	Deliver the final documentation, README and presentation.	4 Dec

1.3 Scope

In scope

- Arabic (MSA, Egyptian) and English; Franco-Arabic evaluated where data allows
- Three classes: Negative, Neutral, Positive
- Public, labelled review and tweet data
- Preprocessing, TF-IDF and n-grams, NB, LR, BiLSTM, Transformer
- Evaluation, MLflow tracking, error analysis, sample predictions
- Optional: Azure ML endpoint and Streamlit demo

Out of scope

- Aspect-based sentiment and emotion detection
- Speech, image or video feedback
- Live social-media crawling and CRM integration
- Collecting private customer data
- Production SLAs and automated retraining

Assumptions and constraints

- Zero budget: free Colab and Kaggle GPUs, open-source tools, and student or DEPI Azure credit for the optional deployment.
- Only public data is used, and no personal data is committed to the repository.
- Base-size Transformers with inputs of up to 128 tokens fit the free GPUs.
- The dates are fixed: kick-off on 19 October and final presentation on 4 December 2026.

1.4 Deliverables

ID	Deliverable (from the brief)	Description	Milestone
D1	Dataset	Unified dataset, label mapping, frozen splits, data card	MS1 · 25 Oct
D2	NLP preprocessing pipeline	Configurable module with unit tests	MS2 · 1 Nov
D3	Classical baseline	TF-IDF with n-grams; Naive Bayes and Logistic Regression	MS2 · 1 Nov
D4	Deep-learning / pretrained model	BiLSTM and a fine-tuned Transformer	MS3 · 15 Nov
D5	Model comparison	Leaderboard with per-language and latency results	MS4 · 22 Nov
D6	MLflow experiment tracking	Every run logged; best model registered	MS4 · 22 Nov
D7	Error analysis	Misclassifications grouped by cause	MS4 · 22 Nov
D8	Sample predictions	Correct and incorrect examples in each language	MS4 · 22 Nov
D9	README	Set-up, reproduction steps, results, limitations	MS6 · 4 Dec
D10	Endpoint and demo *(optional)*	Azure ML endpoint and Streamlit app	MS5 · 29 Nov

1.5 Methodology

The project follows CRISP-DM (Wirth & Hipp, 2000): business understanding, data understanding, data preparation, modelling, evaluation and deployment. These phases run as seven one-week sprints. Each sprint starts with a planning meeting and ends with a short review. Tasks live on a GitHub Projects board, code is merged through reviewed pull requests, and every reported result links to its MLflow run.

1.6 Timeline & Milestones

The project runs from kick-off on **19 October** to the final presentation on **4 December 2026**, in seven one-week sprints (S1–S7). These two dates are fixed; the milestones in between are internal targets.

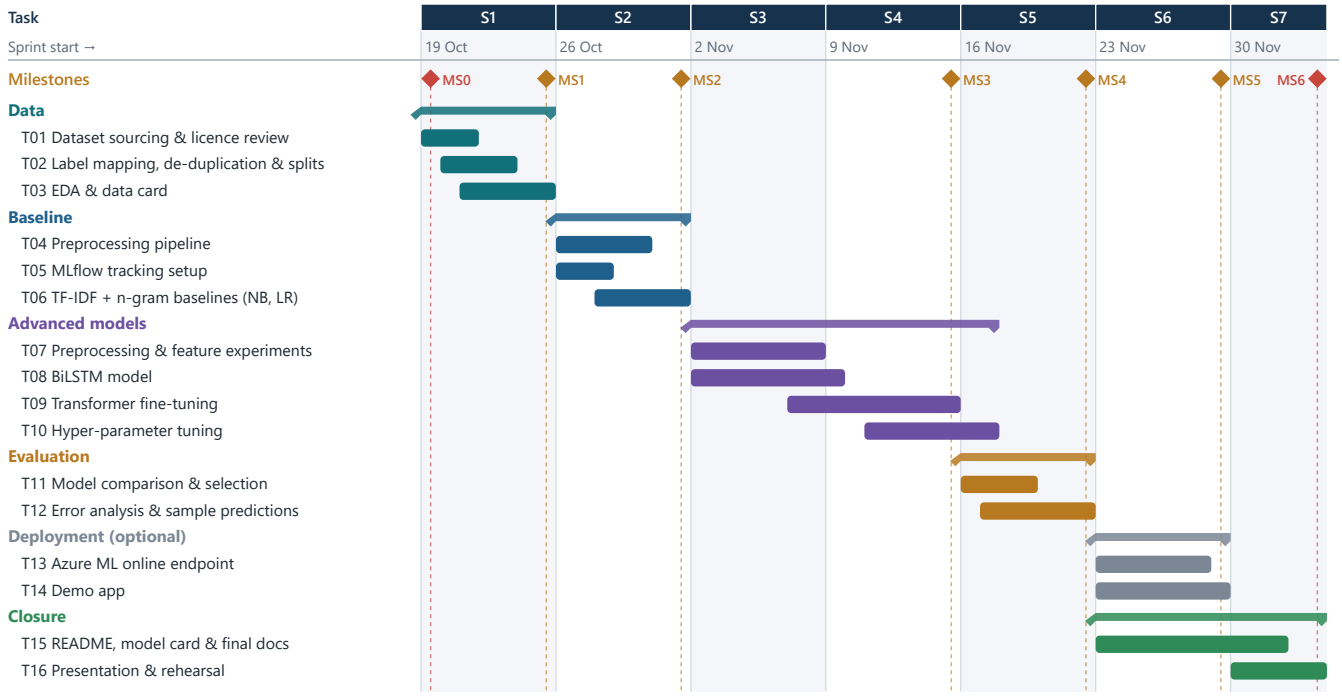


Figure 2 — Project schedule. Bars are coloured by phase; ◆ marks a milestone (red = fixed date).

ID	Milestone	Target date	Exit criteria (evidence)	Type
MS0	**Kick-off: Docs Parts 1–3 + GitHub repo**	19 Oct 2026	Documentation PDF in the repository; repository link registered; LMS assignment submitted	Official DEPI
MS1	**Dataset v1 & EDA**	25 Oct 2026	Unified dataset with frozen train / validation / test splits; data card and EDA reviewed	Internal
MS2	**Baseline ready**	1 Nov 2026	Preprocessing pipeline with tests; NB and LR baselines logged in MLflow	Internal
MS3	**Advanced models trained**	15 Nov 2026	BiLSTM and at least one fine-tuned Transformer evaluated on the test set	Internal
MS4	**Evaluation complete**	22 Nov 2026	Model comparison, confusion matrices, error analysis and sample predictions; best model registered	Internal
MS5	**Deployment & demo (optional)**	29 Nov 2026	Endpoint or demo app returns a label and confidence for single texts and CSV files	Internal
MS6	**Final presentation**	4 Dec 2026	Final documentation, README and presentation delivered	Official DEPI

1.7 Team & Task Distribution

The team has five members:

1. احمد مؤنس احمد عبدالفتاح
2. نوران ناصر جلال سيد
3. محمد محمود محمد عبدالله عياد
4. احمد اشرف سامى فريد
5. رضوه أشرف جمعه مهدي

The work is split into five areas. An owner for each area will be agreed at the first sprint planning meeting and tracked on the GitHub Projects board. Every member takes part in at least one classical and one deep-learning experiment, and in the error analysis.

Work area	Main tasks
Data & EDA	Dataset sourcing, licence checks, label mapping, splits, data card
Preprocessing & classical models	Cleaning, Arabic normalisation, tokenisation, TF-IDF, Naive Bayes, Logistic Regression
Deep learning	BiLSTM, Transformer fine-tuning, hyper-parameter tuning

Work area	Main tasks
MLOps & evaluation	MLflow, evaluation scripts, error analysis, optional Azure deployment
Documentation & presentation	README, reports, final slides

Communication: a weekly planning meeting and sprint review, a team chat for daily updates, and GitHub Issues and pull requests for tasks and code review.

1.8 Risk Management

Risks are reviewed at every sprint review. The level combines likelihood and impact before mitigation.

ID	Risk	Level	Mitigation
R1	Too little labelled Egyptian-dialect customer feedback	High	Combine several public Arabic sources; label a small gold set by hand; report the dialect slice separately
R2	A dataset is withdrawn or has a restrictive licence	High	Check licences in week 1; keep backup sources; commit download scripts, not raw data
R3	Limited free GPU time	High	Base or distilled models, inputs of up to 128 tokens, mixed precision, saved checkpoints
R4	Data leakage: duplicates across splits, or preprocessing fitted on all data	High	De-duplicate before splitting; scikit-learn Pipelines fitted on training data only; frozen test set
R5	Inconsistent labels across sources, and too few Neutral examples	Medium	Written label-mapping policy; manual audit; stratified splits; class weights; macro-F1 as the main metric
R6	Dialect, Franco-Arabic, sarcasm and negation cause errors	Medium	Arabic normalisation, character n-grams, dialect-aware models, error-analysis categories
R7	Uneven availability of team members	Medium	Weekly planning, early warning of delays, shared tasks, buffer in the final week
R8	Azure cost or quota problems for the optional deployment	Low	Smallest suitable instance, deleted after the demo; local Streamlit demo as fallback

1.9 Key Performance Indicators

ID	KPI (model metrics on the frozen test set)	Target
K1	Macro-F1 of the selected model (3 classes)	**≥ 0.75** (minimum 0.70)
K2	Gain over the best classical baseline	**≥ +3 macro-F1 points**
K3	Macro-F1 for each language (Arabic, English)	**≥ 0.65 each**
K4	Latency for one text on CPU	**Baseline ≤ 20 ms; Transformer ≤ 300 ms**
K5	Results traceable to an MLflow run	**100%**
K6	Milestones delivered on time	**All fixed dates; ≥ 90% of internal ones**
K7	Brief deliverables in the repository	**9 / 9**

K1–K3 are provisional and will be reviewed after EDA (MS1). For comparison, a strong Twitter-adapted multilingual model reaches about 0.67 macro-F1 on Arabic and 0.71 on English three-class tweets (Barbieri et al., 2022). Review data is usually easier.

PART 2

2. Literature Review

Related research and existing solutions, what SentiLingo borrows from them, and where it differs.

2.1 Review Method

We searched the ACL Anthology, Google Scholar, arXiv and vendor documentation for work on the sentiment of reviews and social-media text, focusing on Arabic, Egyptian dialect and multilingual models. Every figure quoted below was checked against the original source (October 2026).

2.2 Background and Challenges

Sentiment analysis classifies the opinion expressed in a text. Customer feedback makes this hard: messages are short and informal, often mix praise and complaints, and neutral messages are hard to label consistently. Arabic adds further difficulties:

- rich morphology and inconsistent spelling (e.g. ا/أ, ي/ى);
- dialects such as Egyptian Arabic that differ from MSA in vocabulary and grammar;
- Franco-Arabic and code-switching with English, for which very few labelled resources exist (Hamed et al., 2025);
- sarcasm and label noise, the main sources of the errors that remain in Arabic sentiment models (Abu Farha & Magdy, 2021).

2.3 Classical Approaches

Lexicon-based tools such as VADER score text with a hand-built word list plus rules for negation and intensity. They need no training data but are English-only (Hutto & Gilbert, 2014).

Supervised learning does better:

- On movie reviews, Naive Bayes and SVM classifiers on word features reached 81–83% accuracy and beat keyword lists (Pang et al., 2002).
- Adding word bigrams lifted a Naive Bayes–SVM hybrid to 91.2% (Wang & Manning, 2012).
- For Arabic, an SVM on TF-IDF n-grams scored 0.63 F1 on the four-class Egyptian tweet corpus ASTD (Nabil et al., 2015).
- The best Arabic system at SemEval-2017 was a Naive Bayes model (Rosenthal et al., 2017).

These models are fast, cheap, interpretable and effective with modest data. However, they see no context beyond n-grams, are weak on negation and sarcasm, and break on unseen dialect spellings. This is the baseline required by the brief; SentiLingo adds character n-grams to handle spelling variation.

2.4 Deep Learning and Transformers

LSTMs (Hochreiter & Schmidhuber, 1997) read text as a sequence and can learn effects such as the scope of a negation. They need more data and tuning than linear models.

Pretrained Transformers are first trained on large text corpora and then fine-tuned for the task:

- BERT reached 94.9% on a standard English sentiment benchmark (Devlin et al., 2019).
- XLM-R extends this approach to 100 languages (Conneau et al., 2020).
- XLM-T, a Twitter-adapted version of XLM-R, reached macro-F1 66.9 on Arabic and 70.6 on English three-class tweets (Barbieri et al., 2022).

Arabic-specific models do better on Arabic:

- AraBERT reached 92.6% accuracy on the balanced ASTD, against 80.1% for multilingual BERT (Antoun et al., 2020).

- MARBERT, pretrained on one billion dialect-rich tweets, beat both XLM-R and multilingual BERT on SemEval-2017 Arabic (Abdul-Mageed et al., 2021).

Large language models can classify sentiment without any training. They do well on simple tasks but trail fine-tuned smaller models on harder ones, and they cost much more per prediction (Zhang et al., 2024).

2.5 Arabic and Egyptian-Dialect Studies

The study closest to SentiLingo compared a TF-IDF SVM, an LSTM, multilingual BERT and Arabic BERT models on 40,000 Egyptian tweets (Abu Kwaik et al., 2022). The SVM (0.84 accuracy) matched multilingual BERT (0.83), a Twitter-adapted Arabic BERT was best (0.91) and the LSTM was weakest (0.49); the study used two classes only. A re-implementation of published Arabic methods found that Transformers gave the best F-scores: 0.69 on SemEval-2017 and 0.76 on ASTD (Abu Farha & Magdy, 2021). For code-switched feedback, a corpus of 4,100 Egyptian Arabic–English comments reached an F1 of about 0.93, both with neural models and with a fine-tuned GPT-3.5 (Sherif & Sabty, 2025).

2.6 Existing Solutions

Solution	Type	Arabic support	Strengths	Weaknesses
Azure Language sentiment analysis (Microsoft, 2026)	Cloud API	Generic Arabic	Managed; labels with confidence per sentence and per document	Cannot be customised to the user's data; the feature retires on 31 Mar 2029
Amazon Comprehend (Amazon Web Services, n.d.)	Cloud API	Generic Arabic	Managed; batch jobs; four labels including Mixed	Entity-level sentiment in English only; pay per use
Google Cloud Natural Language (Google Cloud, n.d.)	Cloud API	Generic Arabic	Simple score and magnitude for each sentence	No class labels, so thresholds must be set; no customisation within the API
VADER (Hutto & Gilbert, 2014)	Open-source lexicon	None	Free, fast, no training needed	English only; misses domain and dialect words
Twitter-XLM-R sentiment model (Barbieri et al., 2022)	Open pretrained model	8 languages, including Arabic	One multilingual model, ready to use	Trained on tweets; about 0.67 macro-F1 on Arabic

Table 2.1 — Ready-made options. None of the cloud APIs lists Egyptian Arabic or Franco-Arabic as a supported variety.

2.7 Gap Analysis

#	Gap in existing work	SentiLingo response
G1	Egyptian dialect and Franco-Arabic are poorly covered by ready-made tools	One pipeline for MSA, Egyptian Arabic, English and Franco-Arabic, evaluated per language
G2	Classical and Transformer models are rarely compared on the same data; studies often use one dataset and two classes	The same frozen splits, seeds and metrics for every model family, with three classes
G3	Commercial APIs cannot be customised and can be retired	An owned model that can be retrained on company data
G4	Published results are hard to reproduce	Every run tracked in MLflow, with reproduction steps in the README
G5	LLMs are costly per prediction	Fine-tuned compact models, with latency reported next to accuracy

2.8 Lessons Applied

1. Tune the classical baseline carefully, with word and character n-grams, and keep negation words.
2. Compare a multilingual Transformer with a dialect-aware Arabic model.
3. Report macro-F1 and per-class metrics, with realistic targets: about 0.67–0.71 macro-F1 is already strong for three-class tweets.

PART 3

3. Requirements Gathering

Who SentiLingo serves, what it must do, how well it must do it, and the data and tools it needs.

3.1 Elicitation Approach

Requirements were gathered from five sources:

1. the DEPI project brief, which defines the mandatory scope;
2. the review of existing solutions in Part 2;
3. a team workshop in which members played the stakeholders listed below;
4. inspection of the candidate datasets;
5. guidance from the technical instructor.

Each requirement has an ID and a MoSCoW priority: Must, Should or Could.

3.2 Stakeholders & Their Needs

Stakeholder	Key needs
Customer-experience (CX) managers	Reliable sentiment figures per language and channel; batch analysis; results they can export
Support teams	Negative feedback surfaced first, with a confidence score; works for Egyptian Arabic and Franco-Arabic
Data scientists / ML engineers	A reproducible, configurable pipeline; tracked experiments; easy retraining
IT and security teams	Secure, low-cost deployment; no leakage of customer data
Management	Evidence that the chosen model is accurate and cost-effective
DEPI instructor and evaluators	All brief deliverables, a sound method, and documentation delivered on time

3.3 User Stories

ID	User story	Acceptance criteria
US-01	As a **CX manager**, I want to upload a CSV of feedback and get a sentiment label and confidence for every row.	10,000 rows processed; the output adds label, confidence and language.
US-02	As a **support agent**, I want to paste one message in Arabic, English or Franco-Arabic and see its sentiment.	Result in under 1 second; Arabic shown right-to-left.
US-03	As a **data scientist**, I want every run logged with its parameters, metrics and artefacts.	Each MLflow run holds all brief metrics and a confusion matrix.
US-04	As a **manager**, I want all models compared on accuracy and speed.	A leaderboard shows macro-F1, per-language F1 and latency.
US-05	As an **evaluator**, I want to reproduce the reported results.	Following the README from a fresh clone reproduces the test metrics.

3.4 Functional Requirements

ID	Requirement	Priority
FR-01	Load labelled data from several sources into one schema: text, label, language, source, split.	Must
FR-02	Map each source's labels or star ratings to Negative, Neutral and Positive using a documented policy.	Must
FR-03	Remove duplicates, then create stratified train / validation / test splits with a fixed seed. The test set is frozen.	Must
FR-04	Detect the language or script of each text: Arabic, English or Franco-Arabic.	Should
FR-05	Clean text: mask URLs, mentions, e-mails and phone numbers; strip HTML; handle emojis; lower-case Latin script.	Must

ID	Requirement	Priority
FR-06	Normalise Arabic: remove diacritics and tatweel; unify alef forms (آ إ أ → ا) and ى → ي; reduce repeated letters.	Must
FR-07	Tokenise text: word level for the classical and LSTM models, subwords for Transformers. Negation words are kept.	Must
FR-08	Build TF-IDF features with word and character n-grams.	Must
FR-09	Train Naive Bayes and Logistic Regression with cross-validated tuning.	Must
FR-10	Train a BiLSTM classifier.	Must
FR-11	Fine-tune at least one pretrained Transformer, multilingual and/or Arabic.	Must
FR-12	Report accuracy, precision, recall, F1 (macro and per class) and the confusion matrix, overall and per language.	Must
FR-13	Compare preprocessing strategies and models in one leaderboard.	Must
FR-14	Log every run to MLflow and register the best model.	Must
FR-15	Produce an error analysis and a table of sample predictions.	Must
FR-16	Predict single texts and CSV files, returning a label and a confidence score.	Should
FR-17	Provide a Streamlit demo that shows Arabic right-to-left.	Should
FR-18	Serve the best model from an Azure ML online endpoint (REST, key authentication).	Could
FR-19	Provide a README and a model card.	Must

3.5 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	The selected model meets KPIs K1–K3 (target macro-F1 ≥ 0.75).
NFR-02	Performance	One text is scored on CPU in ≤ 20 ms (baseline) or ≤ 300 ms (Transformer).
NFR-03	Security	No API keys or tokens in the repository; secrets come from environment variables.
NFR-04	Privacy	Personal data (handles, e-mails, phone numbers) is masked, and raw third-party data is never committed.
NFR-05	Usability	A first-time user can score a CSV file in under 2 minutes; Arabic is displayed right-to-left.
NFR-06	Reproducibility	Fixed seeds, pinned dependencies and versioned splits; a re-run reproduces macro-F1 within ±0.01.
NFR-07	Reliability	Empty, emoji-only, mixed-script or very long inputs never crash the pipeline.
NFR-08	Maintainability	Modular code in `src/`, with unit tests for preprocessing and evaluation.
NFR-09	Fairness	Results are reported per language, and the Arabic–English macro-F1 gap stays within 10 points.
NFR-10	Compliance	Datasets and models are used under their licences, and all sources are cited.

3.6 Data Requirements

Label	Definition	English example	Egyptian Arabic example	Franco-Arabic example
Negative	Dissatisfaction, a complaint or criticism	"The order arrived broken."	الطلب وصل مكسور	"el order wesel maksour"
Neutral	No clear sentiment: facts, questions or balanced opinions	"When does the store open?"	المحل بيفتح الساعة كام؟	"el ma7al beyefta7 emta?"
Positive	Satisfaction, praise or a recommendation	"Fast delivery, thank you!"	التوصيل سريع، شكرًا	"el delivery saree3, shokran"

Label mapping: star ratings 1–2 → Negative, 3 → Neutral, 4–5 → Positive. Source labels such as *objective* or *mixed* map to Neutral, and mixed items are flagged so they can be analysed separately.

Candidate dataset	Language	Domain	Size	Licence	Planned use
HARD (Elnagar et al., 2018)	MSA and dialects	Hotel reviews	≈ 374k reviews, rated 1–5	None declared	Main Arabic training data
ASTD (Nabil et al., 2015)	Egyptian and MSA	Tweets	10,006	GPL-2.0	Egyptian-dialect training data
Twitter US Airline Sentiment (CrowdFlower, n.d.)	English	Customer-service tweets	14,640	CC BY-NC-SA 4.0	Main English training data
SemEval-2017 Task 4A (Rosenthal et al., 2017)	Arabic, English	Tweets	9,455 Arabic; 62,617 English	Free use beyond SemEval	Extra three-class training data and a benchmark
UMSAB (Barbieri et al., 2022)	8 languages, including Arabic and English	Tweets	3,030 per language	CC BY 3.0	External multilingual benchmark
JERD — Jumia Egypt Reviews (aliallam13, 2026)	Egyptian, MSA, English, Franco-Arabic	E-commerce reviews	3,809	CC BY-NC-ND 4.0	Held-out test set of real Egyptian feedback

- At least 10,000 labelled texts per core language after cleaning, and every test slice contains all three classes.
- Splits are 70 / 15 / 15, stratified by language and label. The test set is frozen at MS1.
- Duplicates are removed before splitting, licences are recorded for each source, and raw data is not committed.

3.7 Technology Stack

Layer	Tools
Language and environment	Python 3.10+, Jupyter, Google Colab, Kaggle
Data and preprocessing	pandas, Hugging Face Datasets, regular expressions, NLTK, Arabic normalisation utilities
Classical ML	scikit-learn: TF-IDF, Naive Bayes, Logistic Regression, Pipelines
Deep learning	TensorFlow / Keras for the BiLSTM; Hugging Face Transformers on PyTorch
Experiment tracking	MLflow (Zaharia et al., 2018)
Serving and collaboration	Azure Machine Learning, Streamlit; GitHub Issues, Projects and pull requests

3.8 Use-Case Model

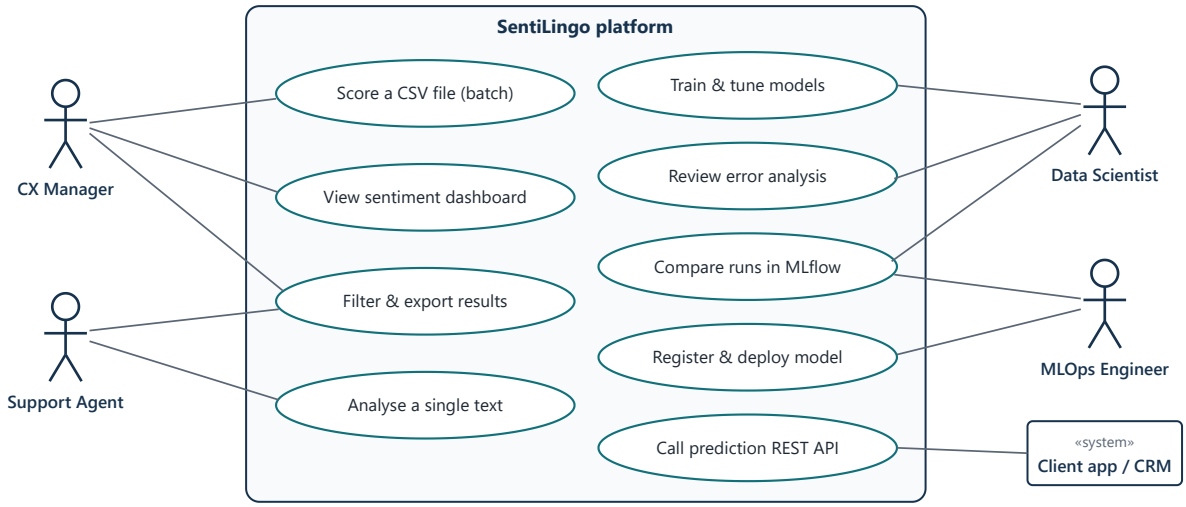


Figure 3 — Use-case diagram: business users (left) consume predictions; technical users and external systems (right) build, deploy and integrate the model.

3.9 Traceability to the Brief

Brief item	Requirements	Deliverable
Dataset	FR-01 – FR-04	D1
Text cleaning, tokenisation	FR-05 – FR-07	D2
TF-IDF, n-grams, Naive Bayes / Logistic Regression	FR-08, FR-09	D3
RNN / LSTM or pretrained Hugging Face Transformer	FR-10, FR-11	D4
Accuracy, precision, recall, F1, confusion matrix	FR-12	D5
Compare preprocessing strategies and models	FR-13	D5
Track experiments using MLflow	FR-14	D6
Error analysis, sample predictions	FR-15	D7, D8
README	FR-19	D9
Optional: Azure ML endpoint	FR-18	D10

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APPENDIX

Appendix A — Glossary

Term	Meaning in this document
Franco-Arabic (Arabizi)	Arabic written in Latin letters, with digits standing for Arabic sounds (e.g. 3 = ع, 7 = ح), as in `el service 7elw`.
Code-switching	Mixing two or more languages or scripts in one message.
MSA / Egyptian Arabic	Modern Standard Arabic, the formal written variety / the everyday Arabic dialect of Egypt.
TF-IDF	Term frequency–inverse document frequency: weights a word by how often it appears in a text, discounted by how common it is across all texts.
N-gram	A sequence of *n* consecutive words or characters, used as a feature.
BiLSTM	A bidirectional Long Short-Term Memory network: a recurrent model that reads text in both directions.
Transformer / fine-tuning	A neural architecture based on self-attention / further training a pretrained model on labelled task data.
Macro-F1	The average of the per-class F1 scores; every class counts equally, so minority classes are not ignored.
Confusion matrix	A table of predicted against true classes, showing which classes the model confuses.
MLflow	An open-source platform for tracking experiments and registering models.
CRISP-DM	A standard data-mining process: business understanding, data understanding, data preparation, modelling, evaluation, deployment.
MoSCoW	A prioritisation scheme: Must, Should, Could, Won't (this time).